

Name: Srinidhi A
ID: COMETFWC053

GATE Question no. 59

Question

Two products are sold from a vending machine, which has two push buttons P_1 and P_2 . When a button is pressed, the price of the corresponding product is displayed in a 7-segment display.

- If no buttons are pressed, '0' is displayed, signifying Rs. 0.
- If only P_1 is pressed, '2' is displayed, signifying Rs. 2.
- If only P_2 is pressed, '5' is displayed, signifying Rs. 5.
- If both P_1 and P_2 are pressed, 'E' is displayed, signifying Error.

Consider:

- push button pressed/not pressed is equivalent to logic 1/0 respectively.
- a segment glowing/not glowing in the display is equivalent to logic 1/0 respectively.

If the segment g and d are considered as functions of P_1 and P_2 , determine the correct logic expressions.

Question Analysis

Inputs: P_1, P_2

Push button pressed = Logic 1 Push button not pressed = Logic 0

Display outputs:

- No button pressed $\rightarrow 0$
- Only P_1 pressed $\rightarrow 2$
- Only P_2 pressed $\rightarrow 5$
- Both pressed $\rightarrow E$

We derive expressions for required segments.

Example:

Segment $g = P_1 + P_2$

Segment $d = c + e$

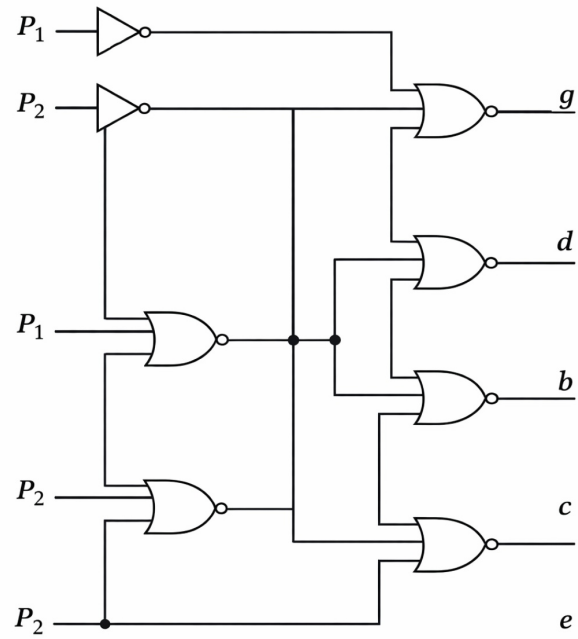
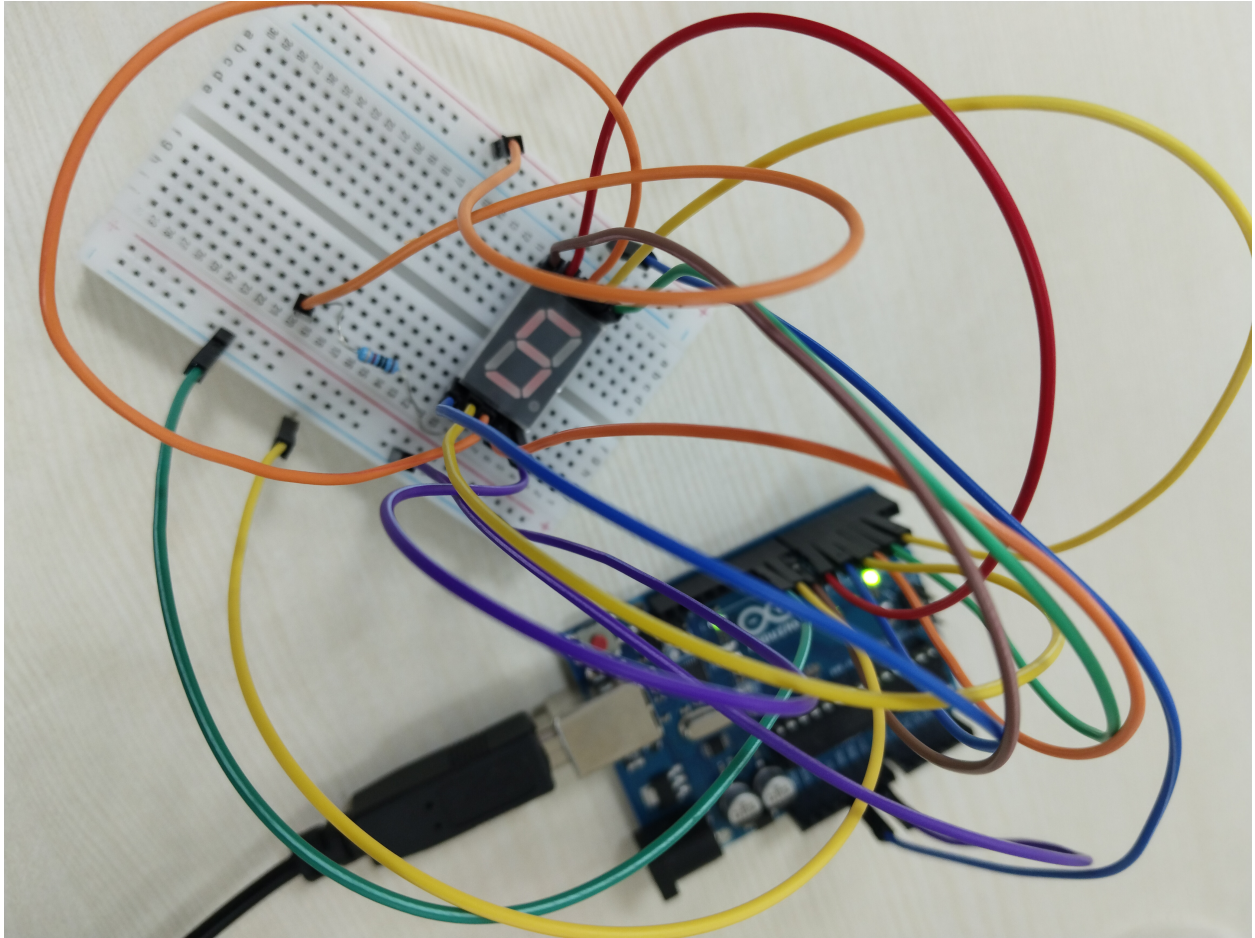


Figure 1: GATE EC 2009 Question 59

Truth Table

P_1	P_2	Display
0	0	0
0	1	5
1	0	2
1	1	E

Hardware Implementation



Required Components

- Arduino UNO
- Breadboard
- 7-segment display
- Push buttons (2)
- Resistors
- Connecting wires
- OTG cable

Pin Connections

- Segment a $\rightarrow$ Pin 2
- Segment b $\rightarrow$ Pin 3
- Segment c $\rightarrow$ Pin 4

- Segment d $\rightarrow$ Pin 5
- Segment e $\rightarrow$ Pin 6
- Segment f $\rightarrow$ Pin 7
- Segment g $\rightarrow$ Pin 8
- $P_1 \rightarrow$ Pin 9
- $P_2 \rightarrow$ Pin 10

Logic Description

The push buttons control the display output. Depending on P_1 and P_2 , the corresponding segments glow to display 0, 2, 5 or E.

Logic expressions are implemented using OR and NOT gates.

Code Uploading Steps

1. Create a Platform IO project.
2. Write the code in `main.cpp` inside `src`.
3. Run the PIO project using command: `pio run`.
4. It compiles the code and creates a `.hex` file.
5. Copy the hex file to ArduinoDroid folder.
6. Connect Arduino UNO to mobile using OTG cable.
7. Upload using "Upload precompiled" option.
8. Observe the output and verify the expression.

Experimental Truth Table

P_1	P_2	Observed Output
0	0	0
0	1	5
1	0	2
1	1	E

Conclusion

The logic for GATE Question 59 was successfully implemented using Arduino UNO and 7-segment display. The experimental results match the theoretical truth table.